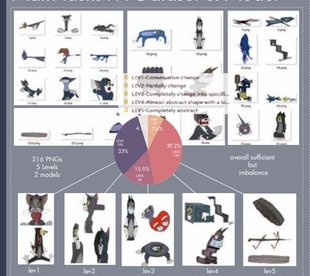
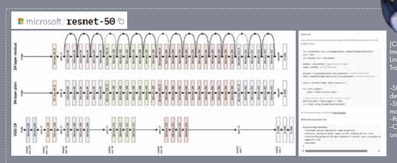
Training&Observation process

At what point does a model stop recognizing the exaggerated Tom as a cat?

STP1 Mar.31st-Apr.3rd

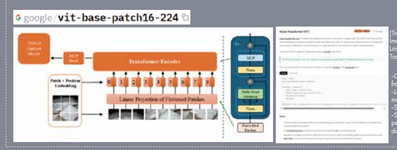
Main tasks1.1-Dataset & Model





[CNN]
Image → Patch Tokens
Linear Projection
Transformer Encoder Blocks

-Stable training of very deep networks
-Strong local texture modeling
-Parameter-efficient
-Computationally efficient on CPU



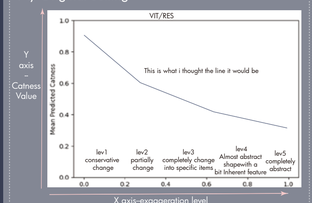
[Transformer-Based]
Image → Patch Tokens
Linear Projection
Transformer Encoder Blocks

-Global receptive field from first layer
-Long-range dependency modeling
-Strong scalability
-State-of-the-art performance with sufficient data

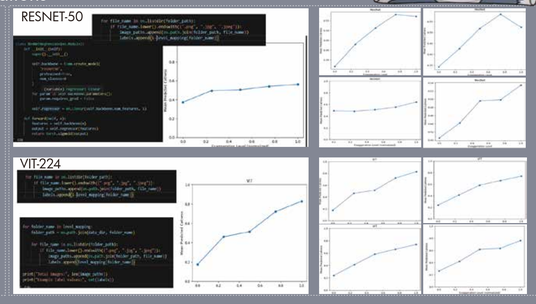
STP2 Apr.5th-Apr.8d

Main tasks 1.2-Preliminary Observations

My Original Setting



This is what i thought the line it would be–The logic I envisioned is actually quite simple.Based on common sense: as Tom becomes more exaggerated—or "levels up"—his anatomical integrity decreases, while his physical abstraction increases. Consequently, the curve should trend downward. **HOWEVER→**

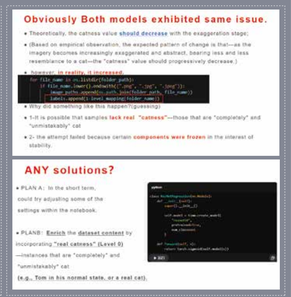


problem in V-B-P-224&Resnet-2026.4.3

catness should be decreasing with the increasement of exaggeration,

but unfortunately it was increasing

maybe it defined catness as sth close to exaggeration (the more abstrat the bigger???)



Conclusion1

Situation?

1.1Dataset&Models
The dataset has been set up; while the overall volume is adequate, the disparities in size between the different groups are excessively large.
All levels are about exaggeration,lack the non-exaggeration as clear comparison

1.2Preliminary Training &Observation
The dataset has been set up; while the overall volume is adequate, the disparities in size between the different groups are excessively large.
ResNet-50 exhibits significantly greater instability than ViT.

At what point does a model stop recognizing the exaggerated Tom as a cat?

For the time being, no definitive conclusion can be drawn regarding this issue, as the underlying patterns themselves have not yet been correctly identified.
However, what can be confirmed is that both models have demonstrated the ability to distinguish between different subsets within the same dataset. It appears that some patterns—specifically those characterized by the principle that "the more exaggerated, the more abstract"—have been uncovered.

STP3 Apr.12th-Apr.14th

Main tasks2.1-Trying Training with Plan A

```
level_mapping = {
    "LEV1-Conservative change": 1.0,
    "LEV2-Partially change": 0.75,
    "LEV3-completely_change_into_specific_items": 0.5,
    "LEV4-Almost_abstract_shape_with_a_bit_inherent_feature": 0.25,
    "LEV5-completely_abstract": 0.0
}
```

Invert the relationship between the exaggeration level and the x-axis coordinate system.

```
levels = sorted(set(res_labels))

res_means = []
vit_means = []

for l in levels:
    res_means.append(res_preds[res_labels == l].mean())
    vit_means.append(vit_preds[vit_labels == l].mean())

plt.figure(figsize=(8,6))
plt.plot(levels, res_means, marker='o', label="ResNet")
plt.plot(levels, vit_means, marker='s', label="ViT")
```

Place the training results of the two models side-by-side to facilitate comparison and observation.

```
plt.xlabel("Exaggeration Level")
plt.ylabel("Mean Predicted Catness")
plt.title("ResNet vs ViT (Frozen Backbone)")
plt.legend()
plt.show()
```

```
self.backbone = timm.create_model(
    "resnet50",
    pretrained=True,
    num_classes=0
)
```

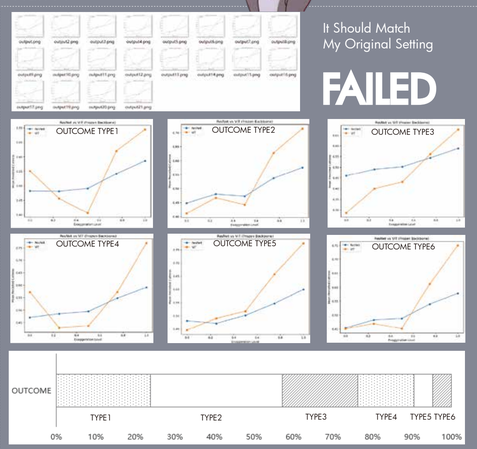
Freeze some layers

```
for param in self.backbone.parameters():
    param.requires_grad = False
```

```
self.backbone = timm.create_model(
    "vit_base_patch16_224",
    pretrained=True,
    num_classes=0
)
```

Freeze some layers

```
for param in self.backbone.parameters():
    param.requires_grad = False
```



It Should Match My Original Setting

FAILED

1- The model overall stick to follow the [more exaggerated the bigger]; moreover, it exhibited significant instability.

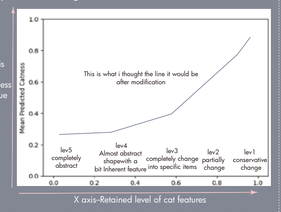
2- At this stage, ViT demonstrated heightened sensitivity and instability.

3-New Orignal Setting&Dataset Modification is needed

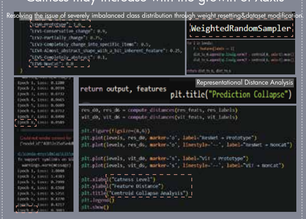
STP4 Apr.15th-Apr.20rd

Main tasks2.2-Trying with Plan B

My New Setting



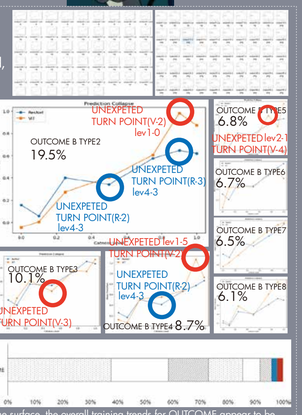
This time, I decided to align with the model's interpretation:The higher the exaggeration level, the smaller the categorical featurevalue; consequently, the corresponding coordinate on the X-axis moves closer to zero.
Catness may increase with the growth of Xaxis



Dataset version B

New Training Process

Dataset increased lev0(no exaggeration)-lev6(its not cat)
I trained it another 50 times; once the situation stabilized, I collected several new charts illustrating the key trends.

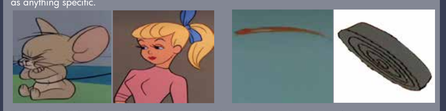


On the surface, the overall training trends for OUTCOME appear to be aligning with the patterns I had hypothesized. However, both VIT and RES exhibited numerous unexpected turning points.
Concurrently, the results displayed in the feature distance plots were mixed—a bittersweet outcome demonstrating that the model is not yet making predictions entirely in accordance with my expectations.
I will address this content in "Conclusion 2," where I will analyze it in conjunction with the pictures from the dataset section.

DEVELOPING

Conclusion2

TURN POINT V-1 is the most frequently occurring "unexpected turn point"—meaning the model often perceives the "LEV6 non-cat" examples as being more cat-like than the "LEV5 completely abstract Tom Cat" examples.
This is actually quite understandable: for the LEV6 dataset, I included images of mice and dogs drawn in a cartoon style—figures that possess standard facial features, limbs, and body shapes. In contrast, the "Tom Cat" examples in LEV5 have become so abstract that they are no longer recognizable as anything specific.

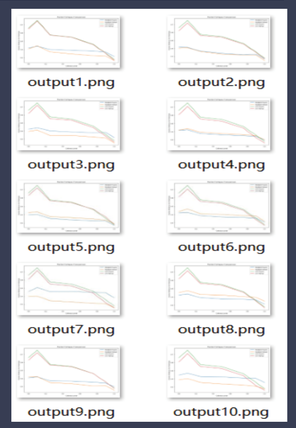
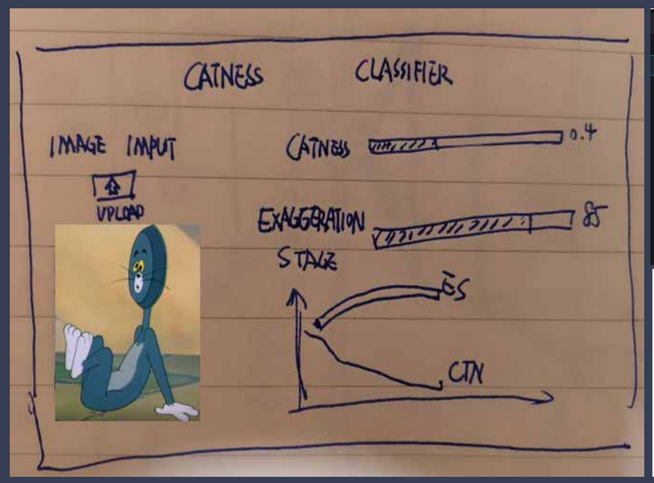
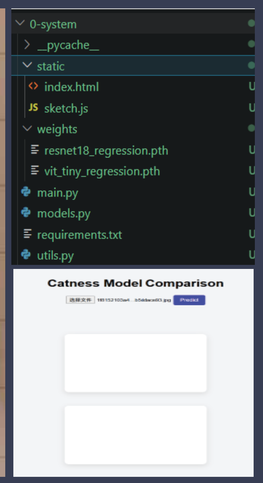
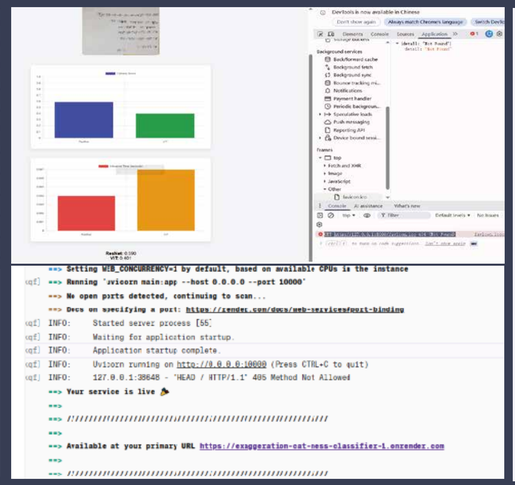


TURN POINT R-2 indicates that the ResNet model frequently perceives LEV4 as being more "cat-like" than LEV3.
I hypothesize that this occurs because, although the fundamental forms of these two datasets have undergone significant changes—with LEV4 being more abstract than LEV3—both still retain relatively similar limbs and facial features; consequently, the model often struggles to distinguish between them. Furthermore, although I categorized LEV3 under "specific item shape," some of those shapes could arguably be considered to reach the level of LEV4's "abstract geometric shape."
I feel this represents an oversight in my classification scheme.

In the feature distance plot:
ResNet's response suddenly became insensitive again; this may suggest that the training depth I have currently applied to the CNN model is insufficient.
As for why the "noncat" values for ViT continued to decrease (converge) within the "distance" metric at LEV5-1—this truly came as a surprise to me. It is possible that as the degree of exaggeration increased, certain facial features and limbs became more pronounced, thereby inadvertently accentuating those specific features.

STP5 Apr.22th-Apr.24th

I should take a plan C but failed. Due to limited time,i choose to just make the online classifier



Plan C Data